

Global Stability of a Hebbian/Anti-Hebbian Network for Principal Subspace Learning

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Biological neural networks self-organize according to local synaptic modifications to produce stable computations. How modifications at the synaptic level give rise to such computations at the network level remains an open question. Pehlevan et al. (2015) proposed a model of a self-organizing neural network with Hebbian and anti-Hebbian synaptic updates that implements an algorithm for principal subspace analysis; however, global stability of the nonlinear synaptic dynamics has not been established. Here, for the case that the feedforward and recurrent weights evolve at the same timescale, we prove global stability of the continuum limit of the synaptic dynamics and show that the dynamics evolve in two phases. In the first phase, the synaptic weights converge to an invariant manifold where the neural filters are orthonormal. In the second phase, the synaptic dynamics follow the gradient flow of a nonconvex potential function whose minima correspond to neural filters that span the principal subspace of the input data.

1 Introduction ---

Biological neural networks self-organize according to local synaptic interactions that produce stable computations at the network level. A challenge in theoretical neuroscience is to link these local interactions to stable network-level computations.

In a seminal work, Oja (1982) proposed a computational model of a neuron as implementing an online algorithm for learning the top principal component of its input data using local, Hebbian synaptic updates. Oja's algorithm thus establishes a link between local, Hebbian interactions at the synaptic level and principal components analysis (PCA) at the neuron level. Moreover, the online (stochastic) algorithm is strikingly stable and

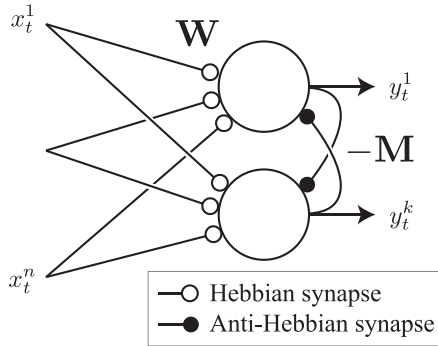


Figure 1: Hebbian/anti-Hebbian network for PSA. Single-layer network with k neurons that receives n inputs. Feedforward Hebbian synapses $\mathbf{W}$ connect the n inputs to the k neurons and recurrent anti-Hebbian synapses; $-\mathbf{M}$ connect the k neurons.

data efficient. It is globally stable with a convergence rate that matches the information-theoretic lower bound (Chou & Wang, 2020), suggesting that neural networks with local synaptic updates may also be relevant in machine learning and neuromorphic computing applications.

In the multichannel setting, much less is known about the stability of neural networks with local synaptic updates. Following Oja's work (Oja, 1982), there were several extensions to principal subspace analysis (PSA) algorithms that can be implemented in neural networks with local Hebbian and anti-Hebbian synaptic updates (Földiák, 1989; Rubner & Tavan, 1989; Rubner & Schulten, 1990; Leen, 1991; Pehlevan et al., 2015, 2018); however, these multichannel networks include recurrent interactions that complicate their analyses, and global stability of the networks' dynamics has not been established. In another line of work, networks with nonlocal synaptic updates for PSA have been proposed and analyzed (Oja & Karhunen, 1983, 1985; Sanger, 1989; Leen et al., 1990; Hornik & Kuan, 1992; Oja, 1992; Plumbley, 1995), but these networks do not explain how local interactions give rise to stable network level computations.

The focus of this work is to analyze the global synaptic dynamics of a multichannel network for principal subspace analysis with Hebbian feedforward synapses and anti-Hebbian recurrent synapses introduced by Pehlevan et al. (2015, 2018) and depicted in Figure 1. Specifically, we show that when the feedforward and recurrent weights evolve on the same timescale, the ordinary differential equation (ODE) associated with the synaptic dynamics is globally stable in the sense that for almost every initialization, the synaptic weights converge to an optimal configuration associated with PSA. We further show that the dynamics evolve in two phases. In the first phase, the synaptic weights converge to an invariant manifold

characterized by orthonormality of the neural filters. In the second phase, the synaptic dynamics follow the gradient flow of a nonconvex potential function. Notably, during the preparation of this letter, we became aware of the preprint, Centorrino et al. (2025) that also analyzes the global stability of the ODE, but in the regime where the recurrent weights evolve on a separate timescale.

2 Hebbian/Anti-Hebbian Network Model

In this section, we first review the Hebbian/anti-Hebbian neural network model for PSA proposed by Pehlevan et al. (2015). A detailed derivation of the network from a so-called similarity matching objective can be found in Pehlevan et al. (2015, 2018). We then introduce the ODE associated with the synaptic dynamics, which will be the focus of our analysis in this work.

Consider a network with k primary neurons that receives $n > k$ inputs, as illustrated in Figure 1. Feedforward synapses $\mathbf{W}$ connect the inputs to primary neurons, and recurrent synapses $-\mathbf{M}$ connect the primary neurons. At each time step $t = 1, 2, \dots$, the network receives an input vector $\mathbf{x}_t = (x_t^1, \dots, x_t^n)$. The network operates on two timescales: a fast neural dynamics timescale and a slow synaptic update timescale.

In the first phase, the neural activities of the k output neurons, which are denoted by the k -dimensional vector $\mathbf{y}_t = (y_t^1, \dots, y_t^k)$, evolve according to fast linear neural dynamics,

$$\dot{\mathbf{y}}_t(\gamma) = \mathbf{W}_t \mathbf{x}_t - \mathbf{M}_t \mathbf{y}_t(\gamma),$$

which converge to $\mathbf{y}_t = \mathbf{F}_t \mathbf{x}_t$, where $\mathbf{W}_t$ and $\mathbf{M}_t$ respectively denote the states of the feedforward and recurrent synaptic weights at time t , and the row vectors of the $k \times n$ matrix $\mathbf{F}_t := \mathbf{M}_t^{-1} \mathbf{W}_t$ are referred to as the “neural filters.” After the neural activities converge, the synaptic weights are updated according to the slow synaptic plasticity rules,

$$\mathbf{W}_{t+1} = \mathbf{W}_t + 2\eta (\mathbf{y}_t \mathbf{x}_t^\top - \mathbf{W}_t), \quad (2.1)$$

$$\mathbf{M}_{t+1} = \mathbf{M}_t + \frac{\eta}{\tau} (\mathbf{y}_t \mathbf{y}_t^\top - \mathbf{M}_t), \quad (2.2)$$

where $\eta > 0$ is the learning rate for the feedforward synapses $\mathbf{W}$ and τ denotes the ratio between the learning rates for the feedforward synapses $\mathbf{W}$ and the lateral synapses $-\mathbf{M}$. The plasticity rule for the feedforward synapses $\mathbf{W}$ includes a term proportional to the product of the pre- and postsynaptic activities, $\mathbf{y}_t \mathbf{x}_t^\top$, so it is referred to as “Hebbian.” The plasticity rule for the lateral synapses $-\mathbf{M}$ includes a term inversely proportional to the product of the pre- and postsynaptic activities, $\mathbf{y}_t \mathbf{y}_t^\top$, so it is referred to as “anti-Hebbian.”

To analyze the stability of their algorithm, Pehlevan et al. (2018) considered the continuum limit of the updates. Formally, when the input data $\{\mathbf{x}_t\}$ are independent and identically distributed samples with zero mean and fixed $n \times n$ covariance matrix $\mathbf{A}$, $\mathbf{M}_0$ is symmetric positive definite, and the step size $\eta > 0$ is infinitesimally small, the synaptic dynamics can be approximated by the ODE:

$$\frac{1}{2} \frac{d\mathbf{W}(t)}{dt} = \mathbf{M}(t)^{-1} \mathbf{W}(t) \mathbf{A} - \mathbf{W}(t). \quad (2.3)$$

$$\tau \frac{d\mathbf{M}(t)}{dt} = \mathbf{M}(t)^{-1} \mathbf{W}(t) \mathbf{A} \mathbf{W}(t)^\top \mathbf{M}(t)^{-1} - \mathbf{M}(t). \quad (2.4)$$

The relationship between the online algorithm, equations 2.1 and 2.2, and the ODE can be made precise for certain time-dependent learning rates under appropriate regularity conditions—for example, the spectrum of $\mathbf{M}$ is uniformly bounded away from zero (Borkar, 1997; Kushner & Clark, 2012).

Pehlevan et al. (2018) proved that every equilibrium point of the ODE corresponds to an eigensubspace of $\mathbf{A}$, and when $\tau \leq \frac{1}{2}$, all linearly stable equilibrium points correspond to the principal (eigen)subspace of $\mathbf{A}$. While their theoretical analysis is informative about the synaptic dynamics near the equilibrium points, it is not informative about the synaptic dynamics away from the equilibrium points, which corresponds to most random initializations. Pehlevan et al. (2015, 2018) provide empirical evidence that both the online algorithm and ODE are globally stable, and they benchmark both against comparable algorithms; however, a proof of global stability for all $\tau \leq \frac{1}{2}$ remains an open problem.

The ODE, equations 2.3 and 2.4, is naturally viewed as the gradient descent-ascent flow (with timescale separation τ) for solving the nonconvex-concave minimax problem,

$$\min_{\mathbf{W}} \max_{\mathbf{M}} f(\mathbf{W}, \mathbf{M}), \quad f(\mathbf{W}, \mathbf{M}) := \text{Tr} \left(-\mathbf{M}^{-1} \mathbf{W} \mathbf{A} \mathbf{W}^\top + \mathbf{W} \mathbf{W}^\top - \frac{1}{2} \mathbf{M}^2 \right), \quad (2.5)$$

where the minimization is over the set of $k \times n$ matrices $\mathbb{R}^{k \times n}$ and the maximization is over the set of $k \times k$ positive-definite matrices $\mathcal{S}_{++}^k$. Analogously, the discrete algorithm equations 2.1 and 2.2, is naturally interpreted as a stochastic gradient descent-ascent algorithm for solving the minimax problem. In general, proving convergence of gradient descent-ascent algorithms for nonconvex-concave minimax problems is challenging. One approach is to rely on a separation of timescales (Borkar, 1997; Lin et al., 2020), and the recent work by Centorrino et al. (2025) applied this approach to prove stability of the ODE, equations 2.3 and 2.4, in the limit $\tau \rightarrow 0$. However, numerical experiments indicate that the optimal convergence rate occurs around $\tau = \frac{1}{2}$ (see Pehlevan et al., 2018, Figure 4), suggesting that a separation of

timescales is unnecessary for proving global convergence and that $\tau = \frac{1}{2}$ may be a parameter of particular interest.

3 Global Stability of the Synaptic Dynamics

For the case $\tau = \frac{1}{2}$, we prove global convergence of the ODE, equations 2.3 and 2.4, to the desired principal subspace. We assume $\mathbf{A}$ is a positive-definite $n \times n$ matrix with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n > 0$ and $k < n$ is fixed. The following theorem is our main result.

Theorem 1. *Let $\mathbf{A} \in S_{++}^n$. For every $\tau > 0$ and $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{D} := \mathbb{R}^{k \times n} \times S_{++}^k$, there exists a unique solution $(\mathbf{W}(t), \mathbf{M}(t))$ of the ODE, equations 2.3 and 2.4, with initial condition $(\mathbf{W}_0, \mathbf{M}_0)$, and $(\mathbf{W}(t), \mathbf{M}(t)) \in \mathcal{D}$ holds for all $t \geq 0$. Moreover, suppose $\lambda_k > \lambda_{k+1}$ and $\tau = \frac{1}{2}$. Then there is a set $\mathcal{Z}$ with Lebesgue measure zero such that if $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{D} \setminus \mathcal{Z}$, then the solution $(\mathbf{W}(t), \mathbf{M}(t))$ converges, as $t \rightarrow \infty$, to the set of equilibrium points $(\mathbf{W}_*, \mathbf{M}_*)$ of the ODE such that the neural filters (i.e., the row vectors of $\mathbf{F}_* := \mathbf{M}_*^{-1} \mathbf{W}_*$) are orthonormal and span the principal subspace of $\mathbf{A}$.*

The next two sections and appendix A are devoted to the proof of theorem 1. In appendix A, we prove the existence and uniqueness of solutions, which ensures that solutions that are initialized in $\mathcal{D}$ remain in $\mathcal{D}$ for all $t \geq 0$. In section 4, we review results from Pehlevan et al. (2018) on local linear stability of equilibrium points, denoted $\mathcal{E}$. Global convergence is proved in section 5. We show that the synaptic weights evolve in two phases. In the first phase, for any initialization outside of a null set $\mathcal{N}$, the synaptic weights first converge to the invariant manifold $\mathcal{O}$ corresponding to orthonormal neural filters; that is, the set of $(\mathbf{W}, \mathbf{M}) \in \mathcal{D}$ such that $\mathbf{F} := \mathbf{M}^{-1} \mathbf{W}$ has orthonormal row vectors (see section 5.1 and Figure 3). This convergence is captured by a convex Lyapunov function $L(\mathbf{W}, \mathbf{M})$. Then, in the second phase, starting on (or near) the invariant manifold $\mathcal{O}$, the synaptic dynamics are approximated by the gradient flow of a nonconvex potential function $V(\mathbf{W})$ (see section 5.2 and Figure 4). As a result, for almost any initialization, the synaptic weights converge to an equilibrium point such that the neural filters correspond to the desired principal subspace projection. A dependency diagram for our main result is shown in Figure 2.

In section 6, we provide empirical evidence that the synaptic weights also evolve in two phases for the discrete online algorithm, equations 2.1 and 2.2. In section 7, we conjecture that theorem 1 can be generalized to hold for all $0 < \tau \leq \frac{1}{2}$.

4 Local Stability of Equilibrium Points

Next, we characterize the equilibrium points of the ODE, equations 2.3 and 2.4, and recall results by Pehlevan et al. (2018) on their linear stability. To

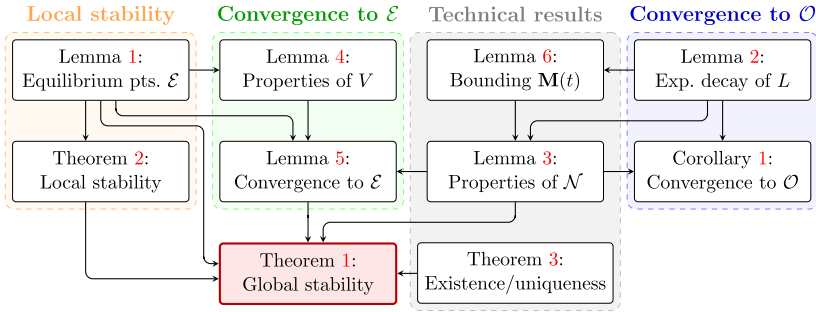


Figure 2: Dependency diagram for proving our main result, theorem 1 (red box). An arrow from a result X to a result Y indicates that result X is referenced in the proof of result Y. Local stability of equilibrium points (orange region) is established in section 4. Convergence of solutions to the invariant manifold $\mathcal{O}$ (blue region) is shown in section 5.1. Convergence starting in or near the invariant manifold to the equilibrium points $\mathcal{E}$ is shown in section 5.2. Finally, global stability of the ODE (red box) is shown in section 5.3. Technical results (gray region) are proved in appendixes A and C.

this end, let

$$\mathcal{E} := \{(\mathbf{W}, \mathbf{M}) \in \mathcal{D} : \mathbf{G}(\mathbf{W}, \mathbf{M}) = \mathbf{0}\}$$

denote the set of equilibrium points, where $\mathbf{G} : \mathcal{D} \mapsto \mathbb{R}^{k \times n} \times S^k$ and where S^k is the set of $k \times k$ symmetric matrices, is the vector field defined by

$$\mathbf{G}(\mathbf{W}, \mathbf{M}) := \left(2\mathbf{M}^{-1}\mathbf{W}\mathbf{A} - 2\mathbf{W}, \frac{1}{\tau}(\mathbf{M}^{-1}\mathbf{W}\mathbf{A}\mathbf{W}^{\top}\mathbf{M}^{-1} - \mathbf{M}) \right). \quad (4.1)$$

4.1 Characterization of Equilibrium points. The following lemma, whose proof is given in appendix B, characterizes the equilibrium points in terms of their singular value decompositions (SVDs):

Lemma 1. Assume $\tau > 0$ and $\mathbf{A} \in S_{++}^n$. A pair $(\mathbf{W}_*, \mathbf{M}_*) \in \mathcal{E}$ if and only if $\mathbf{W}_* = \mathbf{U}\mathbf{S}\mathbf{V}^{\top}$ and $\mathbf{M}_* = \mathbf{U}\mathbf{S}\mathbf{U}^{\top}$, where $\mathbf{U}$ is any $k \times k$ orthogonal matrix, $\mathbf{V}$ is a $n \times k$ matrix whose column vectors are orthonormal eigenvectors of the covariance matrix $\mathbf{A}$, and $\mathbf{S}$ is a $k \times k$ diagonal matrix whose diagonal entries are the eigenvalues of $\mathbf{A}$ corresponding to the column vectors of $\mathbf{V}$.

As a consequence of lemma 1, if $(\mathbf{W}_*, \mathbf{M}_*) \in \mathcal{E}$, then $\mathbf{W}_*$ is full rank (since we assume the eigenvalues of $\mathbf{A}$ are positive) and $\mathbf{F}_* := \mathbf{M}_*^{-1}\mathbf{W}_* = \mathbf{U}\mathbf{V}^{\top}$, where $\mathbf{U}$ is a $k \times k$ orthogonal matrix and the column vectors of $\mathbf{V}$ are orthonormal eigenvectors of $\mathbf{A}$. In other words, the row vectors of $\mathbf{F}_*$ (i.e., the neural filters) are orthonormal and span an eigensubspace of $\mathbf{A}$. In addition, if $(\mathbf{W}_*, \mathbf{M}_*) \in \mathcal{E}$, then $(\mathbf{Q}\mathbf{W}_*, \mathbf{Q}\mathbf{M}_*\mathbf{Q}^{\top}) \in \mathcal{E}$ for every $k \times k$ orthogonal matrix

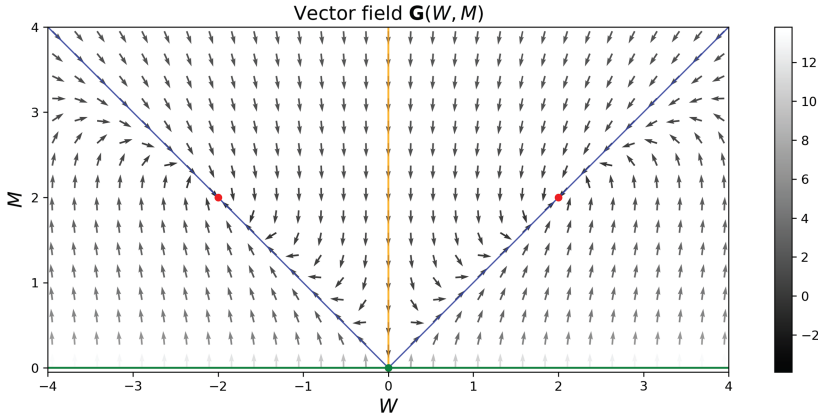


Figure 3: Plot of the vector field $\mathbf{G}(W, M)$ in the case $k = n = 1$ and $\lambda_1 = 2$. The grayscale indicates the logarithm of the vector magnitude. The blue lines denote the set $\mathcal{O}$, the orange vertical line denotes the set $\mathcal{N}$, the two red dots denote the set $\mathcal{E}_0$ (which is equal to $\mathcal{E}$ in this case), and the green line indicates that the line $M = 0$ does not belong to $\mathcal{D} = \mathbb{R} \times (0, \infty)$.

Q. In particular, each equilibrium point is an element of a $\frac{k(k-1)}{2}$ -dimensional manifold of equilibrium points corresponding to an eigen-subspace of $\mathbf{A}$.

4.2 Local Linear Stability Analysis. For our purposes, we say an equilibrium point $(\mathbf{W}_*, \mathbf{M}_*) \in \mathcal{E}$ is linearly stable if all of the eigenvalues of the Jacobian of $\mathbf{G}$ evaluated at $(\mathbf{W}_*, \mathbf{M}_*)$ have a nonpositive real part and it is linearly unstable if it is not linearly stable; that is, at least one of the eigenvalues of the Jacobian of $\mathbf{G}$ evaluated at $(\mathbf{W}_*, \mathbf{M}_*)$ has positive real part. Let

$$\mathcal{E}_0 := \{(\mathbf{W}, \mathbf{M}) \in \mathcal{E} : \text{the rows of } \mathbf{W} \text{ span the principal subspace of } \mathbf{A}\}.$$

The next result follows immediately from Pehlevan et al. (2018, theorem 1).

Theorem 2. Suppose $0 < \tau \leq \frac{1}{2}$ and $\mathbf{A} \in \mathcal{S}_{++}^n$. An equilibrium point $(\mathbf{W}_*, \mathbf{M}_*) \in \mathcal{E}$ is linearly stable if and only if $(\mathbf{W}_*, \mathbf{M}_*) \in \mathcal{E}_0$.

5 Global Convergence Analysis

We now prove our main results on global stability of the ODE, equations 2.3 and 2.4, for $\tau = \frac{1}{2}$, which is fixed throughout this section. Figure 3 shows the vector field $\mathbf{G}(W, M)$ in the scalar case $k = n = 1$. In section 5.1, we show that the neural filters are asymptotically orthonormal. In Figure 3, this corresponds to the initial convergence of trajectories to the blue line. Then, in section 5.2, we show that starting from (or near) an orthonormal

initialization, the ODE, equation 2.3, governing the dynamics of the feed-forward weights $\mathbf{W}$ can be approximated as the gradient flow of a potential function whose minima correspond to the principal subspace. In Figure 3, this corresponds to the convergence of trajectories with initial conditions on (or near) the blue line to the red dots. Finally, in section 5.3, we combine these results to prove theorem 1.

5.1 Asymptotic Orthonormality of the Neural Filters. In this section, we show that for almost every initialization, the solution $(\mathbf{W}(t), \mathbf{M}(t))$ to the ODE, equations 2.3 and 2.4, converges to the following invariant subset of matrices in $\mathcal{D}$ that correspond to orthonormal neural filters $\mathbf{F} = \mathbf{M}^{-1}\mathbf{W}$:

$$\mathcal{O} := \{(\mathbf{W}, \mathbf{M}) \in \mathcal{D} : \mathbf{M}^{-1}\mathbf{W}\mathbf{W}^\top \mathbf{M}^{-1} = \mathbf{I}_k\}. \quad (5.1)$$

To show this convergence, we define the following convex Lyapunov function on $\mathcal{D}$:

$$L(\mathbf{W}, \mathbf{M}) := \|\mathbf{W}\mathbf{W}^\top - \mathbf{M}^2\|^2 = \text{Tr}[(\mathbf{W}\mathbf{W}^\top - \mathbf{M}^2)^2]. \quad (5.2)$$

Note that $L(\mathbf{W}, \mathbf{M})$ is nonnegative everywhere and equal to zero if and only if $(\mathbf{W}, \mathbf{M}) \in \mathcal{O}$.

Lemma 2. Suppose $\mathbf{A} \in \mathcal{S}_{++}^n$ and $(\mathbf{W}(t), \mathbf{M}(t))$ is a solution of the ODE, equations 2.3 and 2.4, with $\tau = \frac{1}{2}$ and initial condition $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{D}$. Then

$$L(\mathbf{W}(t), \mathbf{M}(t)) = L(\mathbf{W}_0, \mathbf{M}_0)e^{-8t}, \quad t \geq 0. \quad (5.3)$$

Proof. First, note that the ODE, equations 2.3 and 2.4, implies

$$\begin{aligned} \frac{d\mathbf{W}(t)}{dt}\mathbf{W}(t)^\top - \frac{d\mathbf{M}(t)}{dt}\mathbf{M}(t) &= 2\mathbf{M}(t)^{-1}\mathbf{W}(t)\mathbf{A}\mathbf{W}(t)^\top - 2\mathbf{W}(t)\mathbf{W}(t)^\top \\ &\quad - [2\mathbf{M}(t)^{-1}\mathbf{W}(t)\mathbf{A}\mathbf{W}(t)^\top \mathbf{M}(t)^{-1}\mathbf{M}(t) - 2\mathbf{M}(t)^2] \\ &= -2[\mathbf{W}(t)\mathbf{W}(t)^\top - \mathbf{M}(t)^2]. \end{aligned}$$

Therefore, by the chain rule, the product rule, the cyclic property of the trace rule, and the previous display,

$$\begin{aligned} \frac{d}{dt}L(\mathbf{W}(t), \mathbf{M}(t)) &= 4\text{Tr}\left[(\mathbf{W}(t)\mathbf{W}(t)^\top - \mathbf{M}(t)^2) \right. \\ &\quad \times \left. \left(\frac{d\mathbf{W}(t)}{dt}\mathbf{W}(t)^\top - \frac{d\mathbf{M}(t)}{dt}\mathbf{M}(t)\right)\right] \\ &= -8L(\mathbf{W}(t), \mathbf{M}(t)). \end{aligned}$$

Solving the differential equation by separation of variables yields equation 5.3. $\square$

To prove that the neural filters are asymptotically orthonormal, we need a technical lemma that states conditions under which the solution $(\mathbf{W}(t), \mathbf{M}(t))$ does not converge to zero. To this end, define the set

$$\mathcal{N} = \{(\mathbf{W}, \mathbf{M}) \in \mathcal{D} : \exists \mathbf{v} \in \mathbb{R}^k, \lambda > 0, \text{ such that } \mathbf{W}^\top \mathbf{v} = \mathbf{0} \text{ and } \mathbf{M}\mathbf{v} = \lambda \mathbf{v}\}. \quad (5.4)$$

Then $\mathcal{N}$ is the set of pairs $(\mathbf{W}, \mathbf{M}) \in \mathcal{D}$ such that $\mathbf{W}\mathbf{W}^\top$ is singular and there is an eigenvector of $\mathbf{M}$ in the null space of $\mathbf{W}\mathbf{W}^\top$. Note that if $\mathbf{W}$ is full rank and $\mathbf{M}$ is positive definite, then $(\mathbf{W}, \mathbf{M}) \notin \mathcal{N}$. The set $\mathcal{N}$ corresponds to the orange vertical line in Figure 3. The following technical lemma establishes that for any initialization not in $\mathcal{N}$, the solution to the ODE, equations 2.3 and 2.4, remains bounded away from zero and infinity. The proof is provided in appendix C.

Lemma 3. *The set $\mathcal{N}$ has Lebesgue measure zero. Suppose $\mathbf{A} \in \mathcal{S}_{++}^n$ and $(\mathbf{W}(t), \mathbf{M}(t))$ is the solution of the ODE, equations 2.3 and 2.4, with $\tau = \frac{1}{2}$ and starting at $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{D}$. If $(\mathbf{W}_0, \mathbf{M}_0) \notin \mathcal{N}$, then $(\mathbf{W}(t), \mathbf{M}(t)) \notin \mathcal{N}$ for all $t \geq 0$ and*

$$\limsup_{t \rightarrow \infty} \{\|\mathbf{M}(t)^{-1}\| + \|\mathbf{W}(t)\|\} < \infty.$$

If $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{N}$, then $(\mathbf{W}(t), \mathbf{M}(t)) \in \mathcal{N}$ for all $t \geq 0$ and

$$\lim_{t \rightarrow \infty} \det(\mathbf{M}(t)) = 0.$$

The following corollary of lemmas 2 and 3 states that almost every solution of the ODE converges exponentially to the invariant manifold $\mathcal{O}$.

Corollary 1. *Suppose $\mathbf{A} \in \mathcal{S}_{++}^n$ and $(\mathbf{W}(t), \mathbf{M}(t))$ is a solution of the ODE, equations 2.3 and 2.4, with $\tau = \frac{1}{2}$ and initial condition $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{D} \setminus \mathcal{N}$. Then $(\mathbf{W}(t), \mathbf{M}(t))$ converges to $\mathcal{O}$ as $t \rightarrow \infty$.*

Proof. By Lemma 3, $K := \sup\{\|\mathbf{M}(t)^{-1}\| : t \geq 0\} < \infty$. Thus,

$$\begin{aligned} \|\mathbf{M}(t)^{-1}\mathbf{W}(t)\mathbf{W}(t)^\top\mathbf{M}(t)^{-1} - \mathbf{I}_k\|^2 &\leq \|\mathbf{M}(t)^{-2}\|^2 \|\mathbf{W}(t)\mathbf{W}(t)^\top - \mathbf{M}(t)^2\|^2 \\ &\leq K^4 e^{-8t} \|\mathbf{W}_0\mathbf{W}_0^\top - \mathbf{M}_0^2\|^2, \end{aligned}$$

where the first inequality is due to the Cauchy-Schwarz inequality, and the second inequality follows from lemma 2. $\square$

5.2 Convergence to Equilibrium Points. Having shown that $(\mathbf{W}(t), \mathbf{M}(t))$ converges to $\mathcal{O}$ as $t \rightarrow \infty$, we now analyze the dynamics when $(\mathbf{W}(t), \mathbf{M}(t))$ is near the set $\mathcal{O}$. To begin, consider the case $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{O}$ so that $(\mathbf{W}(t), \mathbf{M}(t)) \in \mathcal{O}$ for all $t \geq 0$. Then we can rewrite the right-hand side

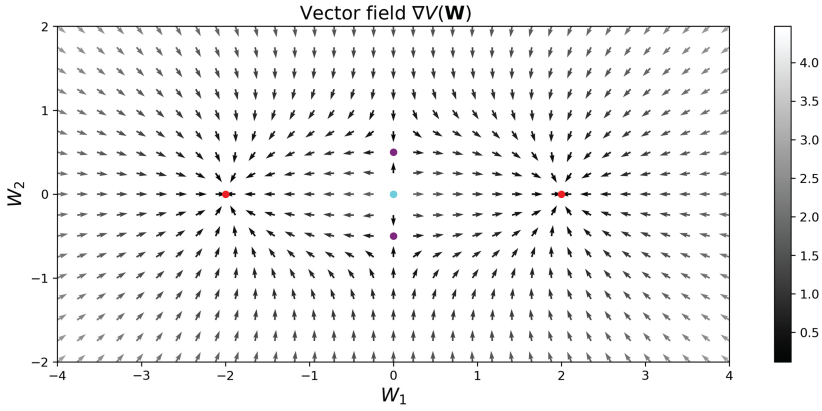


Figure 4: Plot of the vector field $-\nabla V(\mathbf{W})$ in the case $n = 2$, $k = 1$, and $\mathbf{A} = \text{diag}(2, \frac{1}{2})$. The grayscale indicates the vector magnitude. The red dots denote the global minima of V , the purple dots denote the saddle points of V , and the cyan dot at the origin denotes the set $\{\mathbf{W} : \det(\mathbf{W}\mathbf{W}^\top) = 0\}$.

of the ODE, equation 2.3, as a function of $\mathbf{W}(t)$ only:

$$\frac{d\mathbf{W}(t)}{dt} = 2(\mathbf{W}(t)\mathbf{W}(t)^\top)^{-\frac{1}{2}}\mathbf{W}(t)\mathbf{A} - 2\mathbf{W}(t) = -2\nabla V(\mathbf{W}(t)), \quad (5.5)$$

where $V : \mathbb{R}^{k \times n} \mapsto \mathbb{R}$ is the nonconvex potential function

$$V(\mathbf{W}) := f\left(\mathbf{W}, (\mathbf{W}\mathbf{W}^\top)^{\frac{1}{2}}\right) = \text{Tr}\left[-(\mathbf{W}\mathbf{W}^\top)^{-\frac{1}{2}}\mathbf{W}\mathbf{A}\mathbf{W}^\top + \frac{1}{2}\mathbf{W}\mathbf{W}^\top\right], \quad (5.6)$$

where $f(\mathbf{W}, \mathbf{M})$ is defined in equation 2.5. Therefore, we can interpret the $\mathbf{W}(t)$ dynamics on $\mathcal{O}$ as the gradient flow of the potential function V . (Note that V is only differentiable on the subset of full-rank matrices $\mathbf{W}$ in $\mathbb{R}^{k \times n}$.) In Figure 4, we plot the vector field $-\nabla V(\mathbf{W})$ in the case $n = 2$ and $k = 1$ to illustrate the dynamics of $\mathbf{W}(t)$ on the set $\mathcal{O}$.

Lemma 4. Suppose $\mathbf{A} \in \mathcal{S}_{++}^n$. The function V is bounded below. Furthermore, $\nabla V(\mathbf{W})$ exists and satisfies $\nabla V(\mathbf{W}) = 0$ if and only if $\mathbf{W}$ is full rank and $(\mathbf{W}, (\mathbf{W}\mathbf{W}^\top)^{\frac{1}{2}}) \in \mathcal{E}$.

Proof. Let $\mathbf{W} \in \mathbb{R}^{k \times n}$ and let $\mathbf{W} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$ denote its SVD. Then

$$\begin{aligned} V(\mathbf{W}) &= \text{Tr}\left[-\mathbf{V}^\top \mathbf{A} \mathbf{V} \mathbf{S} + \frac{1}{2}\mathbf{S}^2\right] \\ &\geq -\sum_{i=1}^k \sigma_i(\mathbf{A})\sigma_i(\mathbf{S}) + \frac{1}{2}\sum_{i=1}^k \sigma_i^2(\mathbf{S}) \geq -\frac{1}{2}\sum_{i=1}^k \sigma_i^2(\mathbf{A}). \end{aligned}$$

The first equality follows by substituting the SVD of $\mathbf{W}$ into equation 5.6 and using the cyclic property of the trace operator. The first inequality follows from von Neumann's trace inequality (Mirsky, 1975), where $\sigma_i(\mathbf{A})$ and $\sigma_i(\mathbf{S})$ denote the ordered eigenvalues of $\mathbf{A}$ and $\mathbf{S}$, respectively. The second inequality follows from taking derivatives with respect to $\sigma_i(\mathbf{S})$, for $i = 1, \dots, k$, setting the derivatives equal to zero and solving for $\sigma_i(\mathbf{S})$, which yields $\sigma_i(\mathbf{S}) = \sigma_i(\mathbf{A})$. This proves that V is bounded below. Next, suppose $\mathbf{W}$ is full rank and $\nabla V(\mathbf{W}) = 0$. Then $2(\mathbf{W}\mathbf{W}^\top)^{-\frac{1}{2}}\mathbf{W}\mathbf{A}\mathbf{W}^\top = (\mathbf{W}\mathbf{W}^\top)$ and so $\mathbf{V}^\top\mathbf{A}\mathbf{V} = \mathbf{S}$. Since $\mathbf{S}$ is diagonal, it follows that the column vectors of $\mathbf{V}$ are eigenvectors of $\mathbf{A}$ and the diagonal entries of $\mathbf{S}$ are the corresponding eigenvalues of $\mathbf{A}$. Thus, by lemma 1, $(\mathbf{W}, (\mathbf{W}\mathbf{W}^\top)^{\frac{1}{2}}) \in \mathcal{E}$. The converse is readily verified by substitution. $\square$

It follows from LaSalle's invariance principle (LaSalle, 1968), the ODE, equation 5.5, and lemma 4 that when initialized with $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{O}$, the solution $(\mathbf{W}(t), \mathbf{M}(t))$ converges to a fixed point of V , which corresponds to an equilibrium point in $\mathcal{E}$. Next, for the general case $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{D} \setminus \mathcal{N}$, we can rewrite the right-hand side of the ODE, equation 2.3, as follows:

$$\frac{d\mathbf{W}(t)}{dt} = -2\nabla V(\mathbf{W}(t)) + \left[\mathbf{M}(t)^{-1} - (\mathbf{W}(t)\mathbf{W}(t)^\top)^{-\frac{1}{2}} \right] \mathbf{W}(t)\mathbf{A},$$

where we recall that $\mathbf{W}(t)\mathbf{W}(t)^\top$ is nonsingular for all $t \geq 0$ by lemma 3. By the chain rule,

$$\begin{aligned} \frac{dV(\mathbf{W}(t))}{dt} &\leq -2 \|\nabla V(\mathbf{W}(t))\|^2 + \|\nabla V(\mathbf{W}(t))\mathbf{A}\mathbf{W}(t)^\top\| \left\| \mathbf{M}(t)^{-1} \right. \\ &\quad \left. - (\mathbf{W}(t)\mathbf{W}(t)^\top)^{-\frac{1}{2}} \right\|. \end{aligned}$$

We claim that

$$\limsup_{t \rightarrow \infty} \|\nabla V(\mathbf{W}(t))\mathbf{A}\mathbf{W}(t)^\top\| \left\| \mathbf{M}(t)^{-1} - (\mathbf{W}(t)\mathbf{W}(t)^\top)^{-\frac{1}{2}} \right\| = 0.$$

Assuming the claim holds, we have

$$\limsup_{t \rightarrow \infty} \frac{dV(\mathbf{W}(t))}{dt} \leq 0.$$

Therefore, by LaSalle's invariance principle, $\mathbf{W}(t)$ converges to the set of fixed points of V , which correspond to the set of equilibrium points $\mathcal{E}$. We summarize this result in the following lemma and provide a detailed proof in appendix D.

Lemma 5. Suppose $\mathbf{A} \in S_{++}^n$ and $(\mathbf{W}(t), \mathbf{M}(t))$ is a solution to ODE, equations 2.3 and 2.4, with $\tau = \frac{1}{2}$ and initial condition $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{D} \setminus \mathcal{N}$. Then $(\mathbf{W}(t), \mathbf{M}(t))$ converges to the set $\mathcal{E}$ as $t \rightarrow \infty$.

5.3 Proof of Theorem 1. The existence and uniqueness of solutions are shown in theorem 3 in appendix A. We now combine lemma 5 and theorem 2 to prove global convergence. Define the subset

$$\mathcal{U} := \{(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{D} : (\mathbf{W}(t), \mathbf{M}(t)) \text{ converges to } \mathcal{E} \setminus \mathcal{E}_0 \text{ as } t \rightarrow \infty\}$$

of initializations whose trajectories converge to the set of linearly unstable equilibrium points. If $(\mathbf{W}_*, \mathbf{M}_*) \in \mathcal{E} \setminus \mathcal{E}_0$, then by theorem 2, the Jacobian of $\mathbf{G}$ evaluated at $(\mathbf{W}_*, \mathbf{M}_*)$ has an eigenvalue with a positive real part. Therefore, by Potrie and Monzón (2009, proposition 3), the set $\mathcal{U}$ has Lebesgue measure zero. Along with lemma 3, this implies that the set $\mathcal{Z} := \mathcal{N} \cup \mathcal{U}$ also has Lebesgue measure zero. Suppose $(\mathbf{W}(t), \mathbf{M}(t))$ is a solution of the ODE, equations 2.3 and 2.4, with initial condition $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{D} \setminus \mathcal{Z}$. Then by lemma 5 and the definition of $\mathcal{U}$, $(\mathbf{W}(t), \mathbf{M}(t))$ converges to the set $\mathcal{E}_0$ as $t \rightarrow \infty$. Finally, by lemma 1 and the definition of $\mathcal{E}_0$, for every $(\mathbf{W}_*, \mathbf{M}_*) \in \mathcal{E}_0$, the row vectors of $\mathbf{F}_* := \mathbf{M}_*^{-1} \mathbf{W}_*$ are orthonormal and span the principal subspace of $\mathbf{A}$.

6 Comparing the ODE and the Online Algorithm

We now investigate via numerical simulations whether the two-phase convergence observed in the ODE, equations 2.3 and 2.4, also appears in the online algorithm, equations 2.1 and 2.2.

6.1 Setup. We consider a network with $n = 4$ inputs and $k = 2$ neurons. The inputs $\{\mathbf{x}_t\}$ are sampled i.i.d. from a mean zero normal distribution with covariance matrix $\mathbf{A} = \text{diag}(0.5, 0.25, 0.2, 0.05)$. The feedforward weight matrix $\mathbf{W}$ is initialized to have i.i.d. standard normal entries, and the recurrent weight matrix $\mathbf{M}$ is initialized to be diagonal with entries sampled uniformly from $[1, 2]$. The ODE, equations 2.3 and 2.4, is solved using the built-in Mathematica function `NDSolve`. The online algorithm, equations 2.1 and 2.2, is simulated with time-dependent step size $\eta_t = \frac{c_0}{c_1 + t}$, where $c_0, c_1 > 0$ are chosen so that $\eta_1 = 0.001$ and $\sum_{t=1}^{25,000} \eta_t = 8$. To track convergence, we evaluate the Lyapunov functions $L(\mathbf{W}, \mathbf{M})$ and $V_*(\mathbf{W}) = V(\mathbf{W}) - V(\mathbf{W}_*)$, where $(\mathbf{W}_*, \mathbf{M}_*)$ is any stable critical point of the ODE. The ODE and online algorithm are simulated with 100 random initializations.

6.2 Results. Figure 5 illustrates the two-phase convergence for both the ODE and the online algorithm. Panel a shows that $L(\mathbf{W}, \mathbf{M})$ converges exponentially with decay rate e^{-8t} (dotted black line) for the ODE (solid blue

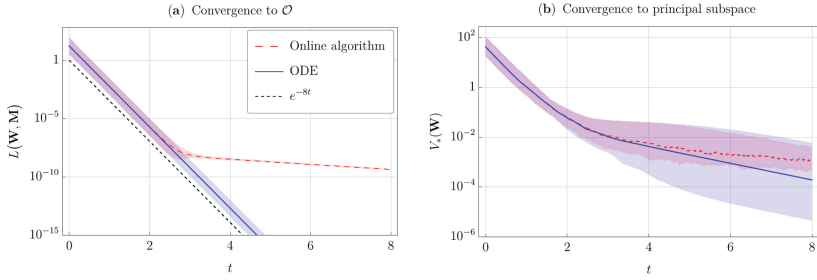


Figure 5: Convergence of the ODE and the online algorithm in a network with $n = 4$ inputs and $k = 2$ neurons. (a) Convergence to the invariant manifold $\mathcal{O}$, measured using the Lyapunov function $L(\mathbf{W}, \mathbf{M})$. (b) Convergence to the principal subspace, measured using $V_*(\mathbf{W}) = V(\mathbf{W}) - V(\mathbf{W}_*)$, where $(\mathbf{W}_*, \mathbf{M}_*)$ is any stable equilibrium point of the ODE. In panels a and b, shaded regions indicate the middle 80th percentile over 100 random initializations; solid and dotted lines denote the median values. Simulation details are in the main text.

line), consistent with lemma 2. For the online algorithm (dashed red line), $L(\mathbf{W}, \mathbf{M})$ initially converges exponentially with decay rate e^{-8t} before leveling off (around $t = 2.5$) due to stochastic fluctuations. Panel b shows convergence to the principal subspace via decay of $V_*(\mathbf{W})$ for the ODE and online algorithm. After the trajectories reach a neighborhood of $\mathcal{O}$ around $t = 2$, the decay rate of $V_*(\mathbf{W})$ for both the ODE and online algorithm continues more gradually as the dynamics of $\mathbf{W}$ are dominated by the gradient $\nabla V(\mathbf{W})$. Overall, the simulations indicate a similar two-phase evolution.

7 Global Stability for General Synaptic Learning Rates

Our analysis focused on the special case $\tau = \frac{1}{2}$, where feedforward and recurrent synapses evolve at equal rates. The choice simplified the structure of the invariant manifold $\mathcal{O}$ and Lyapunov function L , enabling a transparent proof of global convergence. However, the assumption $\tau = \frac{1}{2}$ may appear restrictive and raises the question of whether the same two-phase convergence holds for other $0 < \tau \leq \frac{1}{2}$.

Centorrino et al. (2025) recently established global stability in the limit $\tau \rightarrow 0$, where the recurrent synapses evolve on a timescale separate from that of the feedforward synapses. In this regime, the synaptic dynamics also exhibit a two-phase structure, but with separate timescales. In the first phase, the feedforward weights $\mathbf{W}$ are fixed while the recurrent weights $\mathbf{M}$ evolve toward the invariant manifold $\mathcal{O}_0 = \{(\mathbf{W}, \mathbf{M}) \in \mathcal{D} : \mathbf{M}^{-1} \mathbf{W} \mathbf{A} \mathbf{W}^\top \mathbf{M}^{-1} = \mathbf{M}\}$. As shown in appendix E.1, this convergence can be described by the convex Lyapunov function $L_0(\mathbf{W}, \mathbf{M}) = \|(\mathbf{W} \mathbf{A} \mathbf{W}^\top) - \mathbf{M}^3\|^2$. In the second phase, as shown in Centorrino et al. (2025,

section 5.3), the synaptic matrices $(\mathbf{W}, \mathbf{M})$ evolve within the invariant manifold $\mathcal{O}_0$, and the feedforward synapses $\mathbf{W}$ follow the gradient flow of the nonconvex potential function, $V_0(\mathbf{W}) = -\frac{3}{2}\|(\mathbf{W}\mathbf{A}\mathbf{W}^\top)^{\frac{1}{3}}\|^2 + \|\mathbf{W}\|^2$. Notably, both the invariant manifold $\mathcal{O}_0$ and functions $L_0(\mathbf{W}, \mathbf{M})$ and $V_0(\mathbf{W})$ are distinct from those analyzed here for the case that $\tau = \frac{1}{2}$.

We conjecture that global convergence holds for all $0 < \tau \leq \frac{1}{2}$, with dynamics exhibiting a similar two-phase structure: (1) rapid convergence to a generalized invariant manifold $\mathcal{O}_\tau$ with convergence described by a generalized convex Lyapunov function $L_\tau(\mathbf{W}, \mathbf{M})$, and (2) slower convergence along this manifold to the principal subspace that follows the gradient flow of a generalized nonconvex potential function $V_\tau(\mathbf{W})$. Moreover, we conjecture that for each $0 \leq \tau \leq \frac{1}{2}$, there is a function $\Phi_\tau : \mathbb{R}^{k \times n} \rightarrow \mathcal{S}_+^k$ mapping feedforward weights to recurrent weights and $p_\tau > 0$ such that

- The invariant manifold satisfies $\mathcal{O}_\tau = \{(\mathbf{W}, \mathbf{M}) \in \mathcal{D} : \Phi_\tau(\mathbf{W}) = \mathbf{M}\}$.
- The convex Lyapunov function satisfies $L_\tau(\mathbf{W}, \mathbf{M}) = \|\Phi_\tau(\mathbf{W})^{p_\tau} - \mathbf{M}^{p_\tau}\|^2$.
- The potential function is $V_\tau(\mathbf{W}) = f(\mathbf{W}, \Phi_\tau(\mathbf{W}))$, where $f(\mathbf{W}, \mathbf{M})$ is defined in equation 2.5.

Furthermore, Φ_τ and p_τ satisfy $\Phi_{\frac{1}{2}}(\mathbf{W}) = (\mathbf{W}\mathbf{W}^\top)^{\frac{1}{2}}$ and $p_{\frac{1}{2}} = 2$, and $\lim_{\tau \rightarrow 0} \Phi_\tau(\mathbf{W}) = (\mathbf{W}\mathbf{A}\mathbf{W}^\top)^{\frac{1}{3}}$ and $\lim_{\tau \rightarrow 0} p_\tau = 3$. We anticipate that the conjecture can be proved for τ in a small neighborhood of $\frac{1}{2}$ using an application of the following implicit function theorem for dynamical systems: Persistence of Normally Hyperbolic Invariant Manifolds under Perturbations (Hirsch et al., 1977). The challenge is extending the proof beyond a neighborhood of $\tau = \frac{1}{2}$. In appendix E.2, we provide a numerical method for estimating the invariant manifolds $\mathcal{O}_\tau$ and illustrate the manifolds in the scalar setting $n = k = 1$.

8 Discussion

We proved global convergence of solutions to the ODE, equations 2.3 and 2.4, when $\tau = \frac{1}{2}$. Our analysis revealed a two-phase structure to the convergence: (1) rapid convergence to an invariant set where the neural filters are orthonormal and (2) slower evolution along this manifold following the gradient of a nonconvex potential function whose minima correspond to the principal subspace. This result provides a rigorous link between local Hebbian/anti-Hebbian interactions and stable network-level computations. Practically, the results suggest that initializing synaptic weights on the invariant manifold (e.g., setting $\mathbf{M}_0 = \mathbf{I}_k$ and $\mathbf{W}_0$ to have orthonormal row vectors) could accelerate convergence.

This work is an important step toward proving convergence rate guarantees for the online algorithm analogous to the results established by Chou and Wang (2020) for Oja's PCA model of a neuron. The analysis may also

provide insight into the global dynamics of related algorithms that can be implemented in neural networks with local Hebbian synaptic learning rules for solving nonnegative matrix factorization problems (Pehlevan & Chklovskii, 2014, 2019) and networks with local non-Hebbian synaptic learning rules for solving symmetric generalized eigensubspace problems such as canonical correlation analysis (Lipshutz et al., 2021, 2023).

While the assumption $\tau = \frac{1}{2}$ simplifies the analysis and enables closed-form Lyapunov functions, it is also restrictive and may not be biologically realistic. Future work should extend the proof of global stability to the range $0 < \tau \leq \frac{1}{2}$, as supported by numerical evidence (see section 7 and appendix E) and analytical results for the regime $\tau \rightarrow 0$ (Centorrino et al., 2025).

Appendix A: Existence and Uniqueness of Solutions

In this appendix, we prove the existence and uniqueness of solutions to the ODE, for any $\tau > 0$. A solution of the ODE is a continuously differentiable function $t \mapsto (\mathbf{W}(t), \mathbf{M}(t))$ from $[0, \infty)$ to $\mathcal{D}$ whose derivative satisfies equations 2.3 and 2.4. Recall that $\|\cdot\|$ denotes the Frobenius norm.

Theorem 3. *Suppose $\mathbf{A} \in S_{++}^n$. For any $\tau > 0$ and $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{D}$, there exists a unique solution $(\mathbf{W}(t), \mathbf{M}(t))$ to the ODE equations 2.3–2.4 with initial condition $(\mathbf{W}_0, \mathbf{M}_0)$. Moreover, $(\mathbf{W}(t), \mathbf{M}(t)) \in \mathcal{D}$ for all $t \geq 0$.*

Proof. Recall the vector field $\mathbf{G} : \mathcal{D} \rightarrow \mathbb{R}^{k \times n} \times S^k$ defined in equation 4.1, where S^k is the vector space of $k \times k$ symmetric matrices. For each $K < \infty$, define the set

$$\mathcal{D}_K := \{(\mathbf{W}, \mathbf{M}) \in \mathcal{D} : \|\mathbf{M}^{-1}\| + \|\mathbf{W}\| < K\}.$$

Since the vector field $\mathbf{G}$ is continuously differentiable on $\mathcal{D}$ and, for $K < \infty$, the closure of $\mathcal{D}_K$ is compact and contained in $\mathcal{D}$, it follows that $\mathbf{G}$ is uniformly Lipschitz continuous on the closure of $\mathcal{D}_K$.

Let $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{D}$, and choose $K < \infty$ sufficiently large that $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{D}_K$. There exists a unique solution $(\mathbf{W}(t), \mathbf{M}(t))$ of the ODE, equations 2.3 and 2.4, on the interval $[0, T_K)$, where

$$T_K := \inf \{t \geq 0 : (\mathbf{W}(t), \mathbf{M}(t)) \notin \mathcal{D}_K\}$$

is the first time $(\mathbf{W}(t), \mathbf{M}(t))$ exits the set $\mathcal{D}_K$. Notably, by the definition of $\mathcal{D}_K$, $\mathbf{M}(t)$ remains symmetric positive-definite for all $t \in [0, T_K)$. Let $T_\infty := \lim_{K \rightarrow \infty} T_K$. It follows that there exists a unique solution of the ODE, equations 2.3 and 2.4, with initial condition $(\mathbf{W}_0, \mathbf{M}_0)$ on the interval $[0, T_\infty)$, and $(\mathbf{W}(t), \mathbf{M}(t)) \in \mathcal{D}$ for all $t \in [0, T_\infty)$.

We are left to show that $T_\infty = \infty$. Given any $T < \infty$, define

$$K := \max \left\{ \frac{4\sqrt{k}}{m} e^{T/\tau}, 4\|\mathbf{W}_0\| \exp \left[2 \left(\frac{\sqrt{k}}{m} \|\mathbf{A}\| e^{T/\tau} + 1 \right) T \right] \right\} < \infty.$$

We prove that $T_K > T$, which will complete the proof. Let $\mathbf{v} \in \mathbb{R}^k$ be an arbitrary unit vector and define $a(t) := \mathbf{v}^\top \mathbf{M}(t) \mathbf{v}$ for all $t \in [0, T_K)$. By the ODE, equation 2.4,

$$\tau \frac{da(t)}{dt} = \mathbf{v}^\top \mathbf{M}(t)^{-1} \mathbf{W}(t) \mathbf{A} \mathbf{W}(t)^\top \mathbf{M}(t)^{-1} \mathbf{v} - a(t) \geq -a(t),$$

and so $a(t) \geq a(0)e^{-t/\tau}$ for all $t \in [0, T_K)$. Since this holds for all unit vectors $\mathbf{v} \in \mathbb{R}^k$, we have

$$\sigma_{\min}(\mathbf{M}(t)) \geq m e^{-t/\tau} > 0, \quad t \in [0, T_K), \quad (\text{A.1})$$

where $m := \sigma_{\min}(\mathbf{M}_0)$ and $\sigma_{\min}(\mathbf{M}) > 0$ denotes the smallest eigenvalue of $\mathbf{M}$. Therefore,

$$\|\mathbf{M}(t)^{-1}\| = \sqrt{\text{Tr}(\mathbf{M}(t)^{-2})} \leq \frac{\sqrt{k}}{\sigma_{\min}(\mathbf{M}(t))} \leq \frac{\sqrt{k}}{m} e^{t/\tau}, \quad t \in [0, T_K). \quad (\text{A.2})$$

Next, by the ODE, equation 2.3, for all $t \in [0, T_K)$,

$$\begin{aligned} \left\| \frac{d\mathbf{W}(t)}{dt} \right\| &\leq 2\|\mathbf{M}(t)^{-1}\| \|\mathbf{W}(t)\| \|\mathbf{A}\| + 2\|\mathbf{W}(t)\| \\ &\leq 2 \left(\frac{\sqrt{k}}{m} \|\mathbf{A}\| e^{t/\tau} + 1 \right) \|\mathbf{W}(t)\|. \end{aligned}$$

By Gronwall's inequality,

$$\|\mathbf{W}(t)\| \leq \|\mathbf{W}_0\| \exp \left[2 \left(\frac{\sqrt{k}}{m} \|\mathbf{A}\| e^{t/\tau} + 1 \right) t \right], \quad t \in [0, T_K). \quad (\text{A.3})$$

By the definition of K and the bounds, equations A.2 and A.3, $\|\mathbf{M}(t)^{-1}\| + \|\mathbf{W}(t)\| < \frac{K}{2}$ for all $t < \min(T, T_K)$. The continuity of solutions ensures that $T_K > \min(T, T_K)$, and so $T_K > T$. $\square$

Appendix B: Characterizing the Critical Points

In this section, we prove lemma 1, which characterizes the critical points of the ODE, equations 2.3 and 2.4.

Proof of Lemma 1. First suppose $(\mathbf{W}_*, \mathbf{M}_*) \in \mathcal{E}$. Setting the derivative in equation 2.4 to zero, we see that $(\mathbf{W}_*, \mathbf{M}_*)$ satisfy

$$\mathbf{M}_*^{-1} \mathbf{W}_* \mathbf{A} \mathbf{W}_*^\top \mathbf{M}_*^{-1} = \mathbf{M}_*.$$

After multiplying both sides of the equality on the left and right by $\mathbf{M}_*$, we obtain the relation $\mathbf{M}_*^3 = \mathbf{W}_* \mathbf{A} \mathbf{W}_*^\top$. Taking cube roots yields $\mathbf{M}_* = (\mathbf{W}_* \mathbf{A} \mathbf{W}_*^\top)^{\frac{1}{3}}$. Next, setting the derivative in equation 2.3 to zero, we obtain

$$\mathbf{M}_*^{-1} \mathbf{W}_* \mathbf{A} = \mathbf{W}_*.$$

After right-multiplying by $\mathbf{W}_*^\top$ on both sides of the equality and substituting in with $\mathbf{M}_*^{-1} = (\mathbf{W}_* \mathbf{A} \mathbf{W}_*^\top)^{-\frac{1}{3}}$, we obtain $(\mathbf{W}_* \mathbf{A} \mathbf{W}_*^\top)^{\frac{2}{3}} = \mathbf{W}_* \mathbf{W}_*^\top$. Substituting in with the SVD $\mathbf{W}_* = \mathbf{U} \mathbf{S} \mathbf{V}^\top$ yields $\mathbf{V}^\top \mathbf{A} \mathbf{V} = \mathbf{S}$. Since $\mathbf{S}$ is diagonal, it follows that the column vectors of $\mathbf{V}$ are eigenvectors of $\mathbf{A}$ and the diagonal elements of $\mathbf{S}$ are the corresponding eigenvalues. Finally, we see that $\mathbf{M}_* = \mathbf{U} \mathbf{S} \mathbf{U}^\top$. On the other hand, if $\mathbf{W}_*$ and $\mathbf{M}_*$ are as in the statement of the lemma, then it is readily verified by substitution that $\mathbf{G}(\mathbf{W}_*, \mathbf{M}_*) = \mathbf{0}$. $\square$

Appendix C: Bounding the Synaptic Weights

In this section, we prove lemma 3, which states that the set $\mathcal{N}$ of initial conditions with divergent trajectories, has Lebesgue measure zero. Through the remainder of the appendixes we fix $\tau = \frac{1}{2}$. We first need the following lemma.

Lemma 6. Suppose $\mathbf{A} \in \mathcal{S}_{++}^n$ and $(\mathbf{W}(t), \mathbf{M}(t))$ is a solution of the ODE, equations 2.3 and 2.4, with initial condition $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{D} \setminus \mathcal{N}$. Then $\liminf_{t \rightarrow \infty} \det(\mathbf{M}(t)) > 0$ and $\limsup_{t \rightarrow \infty} \|\mathbf{M}(t)\| < \infty$.

Proof. For a proof by contradiction, suppose $\liminf_{t \rightarrow \infty} \det(\mathbf{M}(t)) = 0$. Then, along with lemma 2, this implies there is a sequence $\{t_j\}$ with $t_j \rightarrow \infty$ as $j \rightarrow \infty$ such that

$$\lim_{j \rightarrow \infty} \det(\mathbf{M}(t_j)^2) = \lim_{j \rightarrow \infty} \det(\mathbf{W}(t_j) \mathbf{W}(t_j)^\top) = 0,$$

and for each $j = 1, 2, \dots$,

$$-\infty < \frac{d}{dt} \log \det(\mathbf{W}(t_j) \mathbf{W}(t_j)^\top) < 0.$$

Since $(\mathbf{W}, \mathbf{M}) \notin \mathcal{N}$, we can choose j sufficiently large such that

$$\frac{k}{\sigma_{\min}(\mathbf{A})} < \text{Tr}[\mathbf{M}(t_j)^{-1}] < \infty. \quad (\text{C.1})$$

However, by the ODE, equation 2.3,

$$\begin{aligned}
 \frac{d}{dt} \log(\det(\mathbf{W}(t_j)\mathbf{W}(t_j)^\top)) &= \text{Tr} \left[(\mathbf{W}(t_j)\mathbf{W}(t_j)^\top)^{-1} \frac{d\mathbf{W}(t_j)\mathbf{W}(t_j)^\top}{dt} \right] \\
 &= 4\text{Tr} \left[(\mathbf{W}(t_j)\mathbf{W}(t_j)^\top)^{-1} \mathbf{M}(t_j)^{-1} \right. \\
 &\quad \left. \times \mathbf{W}(t_j)\mathbf{A}\mathbf{W}(t_j)^\top - \mathbf{I}_k \right] \\
 &\geq 4\text{Tr} \left[\sigma_{\min}(\mathbf{A}) (\mathbf{W}(t_j)\mathbf{W}(t_j)^\top)^{-1} \right. \\
 &\quad \left. \times \mathbf{M}(t_j)^{-1} \mathbf{W}(t_j)\mathbf{W}(t_j)^\top - \mathbf{I}_k \right], \\
 &> 0,
 \end{aligned}$$

a contradiction. Therefore, $\liminf_{t \rightarrow \infty} \det(\mathbf{M}(t)) > 0$.

We are left to show that $\limsup_{t \rightarrow \infty} \|\mathbf{M}(t)\| < \infty$. Again, for a proof by contradiction, suppose $\limsup_{t \rightarrow \infty} \|\mathbf{M}(t)\| = \infty$. By the previous result, $K = \sup_{t \geq 0} \|\mathbf{M}(t)^{-1}\| < \infty$. Thus, there exists $t > 0$ sufficiently large such that

$$\frac{d\|\mathbf{M}(t)\|^2}{dt} > 0 \quad (\text{C.2})$$

and

$$-\infty < \text{Tr} [\sigma_{\max}(\mathbf{A})\mathbf{M}(t) - \mathbf{M}(t)^2] + 4KL(\mathbf{W}(0), \mathbf{M}(0))e^{-4t} \sigma_{\max}(\mathbf{A}) < 0.$$

By the ODE, equation 2.4, and lemma 2,

$$\begin{aligned}
 \frac{d\|\mathbf{M}(t)\|^2}{dt} &= 4\text{Tr} [\mathbf{M}(t)^{-1}\mathbf{W}(t)\mathbf{A}\mathbf{W}(t)^\top - \mathbf{M}(t)^2] \\
 &\leq 4\text{Tr} [\sigma_{\max}(\mathbf{A})\mathbf{M}(t)^{-1}\mathbf{W}(t)\mathbf{W}(t)^\top - \mathbf{M}(t)^2] \\
 &\leq 4\text{Tr} [\sigma_{\max}(\mathbf{A})\mathbf{M}(t) - \mathbf{M}(t)^2] + 4KL(\mathbf{W}(0), \mathbf{M}(0))e^{-4t} \sigma_{\max}(\mathbf{A}) \\
 &< 0.
 \end{aligned}$$

This contradicts equation C.2. Therefore, $\limsup_{t \rightarrow \infty} \|\mathbf{M}(t)\| < \infty$. $\square$

Proof of Lemma 3. The fact that $\mathcal{N}$ has Lebesgue measure zero follows immediately because $\mathcal{N} \subset \{(\mathbf{W}, \mathbf{M}) \in \mathcal{D} : \det(\mathbf{W}\mathbf{W}^\top) = 0\}$, $\det(\cdot)$ is a nonzero polynomial, and the vanishing set of a nonzero polynomial has Lebesgue measure zero (Caron & Traynor, 2005).

Let $(\mathbf{W}(t), \mathbf{M}(t))$ be a solution of the ODE, equations 2.3 and 2.4, with initial condition $(\mathbf{W}_0, \mathbf{M}_0)$. In the case $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{D} \setminus \mathcal{N}$, lemmas 2 and 6 imply that $\|\mathbf{M}(t)^{-1}\|$ and $\|\mathbf{W}(t)\|$ are uniformly bounded in t . On the other

hand, suppose $(\mathbf{W}_0, \mathbf{M}_0) \in \mathcal{N}$, and $(\mathbf{W}(t), \mathbf{M}(t))$ is the solution to equations 2.3 and 2.4, with initial condition $(\mathbf{W}_0, \mathbf{M}_0)$. Then there exists $t > 0$, $\mathbf{v} \in \mathbb{R}^k$ and $\lambda \in \mathbb{R}$ such that $\mathbf{W}(t)^\top \mathbf{v} = 0$ and $\mathbf{M}(t)\mathbf{v} = \lambda \mathbf{v}$. Then

$$\begin{aligned} \frac{d\mathbf{v}^\top \mathbf{W}(t)}{dt} &= 2(\mathbf{v}^\top \mathbf{W}(t)\mathbf{M}(t)^{-1}\mathbf{A} - \mathbf{v}^\top \mathbf{W}(t)) = 0 \\ \frac{d\mathbf{M}(t)\mathbf{v}}{dt} &= 2(\mathbf{M}(t)^{-1}\mathbf{W}(t)\mathbf{A}\mathbf{W}(t)^\top \mathbf{M}(t)^{-1}\mathbf{v} - \mathbf{M}(t)\mathbf{v}(t)) = -2\lambda \mathbf{v}. \end{aligned}$$

It follows that for $t > 0$, $\mathbf{W}(t)^\top \mathbf{v} = 0$ and $\mathbf{M}(t)\mathbf{v} = \lambda e^{-2t} \mathbf{v}$, $(\mathbf{W}(t), \mathbf{M}(t)) \in \mathcal{N}$. Therefore, for each $t > 0$, $\mathbf{v}$ is an eigenvector of $\mathbf{M}(t)$ with eigenvalue λe^{-2t} . It follows that $\lim_{t \rightarrow \infty} \det(\mathbf{M}(t)) = 0$. $\square$

Appendix D: Convergence of Neural Filters to an Eigen Subspace

In this section, we prove lemma 5.

Proof of Lemma 5. Suppose $(\mathbf{W}(t), \mathbf{M}(t))$ is a solution to ODE, equations 2.3 and 2.4. By the definition of $V(\mathbf{W})$ in equation 5.6 and the ODE, equation 2.3, we have (suppressing the dependence on t)

$$\begin{aligned} \text{Tr} \left[\nabla_{\mathbf{W}}(V(\mathbf{W})) \frac{d\mathbf{W}^\top}{dt} \right] &= 2\text{Tr} \left[\left(\mathbf{W} - (\mathbf{W}\mathbf{W}^\top)^{-\frac{1}{2}} \mathbf{W}\mathbf{A} \right) (\mathbf{A}\mathbf{W}^\top \mathbf{M}^{-1} - \mathbf{W}) \right] \\ &= 2\text{Tr} \left[\left(\mathbf{W} - (\mathbf{W}\mathbf{W}^\top)^{-\frac{1}{2}} \mathbf{W}\mathbf{A} \right) \left(\mathbf{A}\mathbf{W}^\top (\mathbf{W}\mathbf{W}^\top)^{-\frac{1}{2}} - \mathbf{W} \right) \right] \\ &\quad + \text{Tr} \left[\left(\mathbf{W} - (\mathbf{W}\mathbf{W}^\top)^{-\frac{1}{2}} \mathbf{W}\mathbf{A} \right) \mathbf{A}\mathbf{W}^\top \right. \\ &\quad \left. \times \left(\mathbf{M}^{-1} - (\mathbf{W}\mathbf{W}^\top)^{-\frac{1}{2}} \right) \right] \\ &= -2 \left\| \nabla_{\mathbf{W}}(V(\mathbf{W})) \right\|^2 \\ &\quad + \text{Tr} \left[\left(\mathbf{W} - (\mathbf{W}\mathbf{W}^\top)^{-\frac{1}{2}} \mathbf{W}\mathbf{A} \right) \mathbf{A}\mathbf{W}^\top \right. \\ &\quad \left. \times \left(\mathbf{M}^{-1} - (\mathbf{W}\mathbf{W}^\top)^{-\frac{1}{2}} \right) \right]. \end{aligned}$$

By lemma 3, $\mathbf{W}(t) - (\mathbf{W}(t)\mathbf{W}(t)^\top)^{-\frac{1}{2}} \mathbf{W}(t)\mathbf{A}$ is uniformly bounded in t . By lemmas 2 and 3,

$$\begin{aligned} \limsup_{t \rightarrow \infty} \left\| \mathbf{M}(t)^{-1} - (\mathbf{W}(t)\mathbf{W}(t)^\top)^{-\frac{1}{2}} \right\|^2 \\ \leq \limsup_{t \rightarrow \infty} \left\| \mathbf{M}(t)^{-1} \right\|^2 \left\| (\mathbf{W}(t)\mathbf{W}(t)^\top)^{\frac{1}{2}} \right\|^2 \left\| \mathbf{M}(t) - (\mathbf{W}(t)\mathbf{W}(t)^\top)^{\frac{1}{2}} \right\|^2 = 0. \end{aligned}$$

Therefore, along with LaSalle's invariance principle and lemma 4, this implies that $(\mathbf{W}(t), \mathbf{M}(t))$ converges to the set $\mathcal{E}$ as $t \rightarrow \infty$. This completes the proof. $\square$

Appendix E: Stability of the ODE for General Timescales

In section 7, we conjectured that the global convergence result proved in the main text for $\tau = \frac{1}{2}$ (theorem 1) extends to all $0 < \tau \leq \frac{1}{2}$, and that the dynamics retain the same two-phase structure: (1) rapid convergence toward a τ -dependent invariant manifold $\mathcal{O}_\tau$, captured by a convex Lyapunov function L_τ , and (2) slow evolution along $\mathcal{O}_\tau$ governed by the gradient flow of a τ -dependent nonconvex potential V_τ , whose minima correspond to the principal subspace. Here we provide some limited empirical evidence in support of our conjecture.

E.1 Convex Lyapunov Function for the Case $\tau \rightarrow 0$. In the $\tau \rightarrow 0$ regime, the recurrent weights first converge to the invariant manifold $\mathcal{O}_0 = \{(\mathbf{W}, \mathbf{M}) \in \mathcal{D} : \mathbf{M}^{-1} \mathbf{W} \mathbf{A} \mathbf{W}^\top \mathbf{M}^{-1} = \mathbf{M}\}$ while the feedforward weights remain fixed, after which both weights evolve within the invariant manifold. Here, we sketch out an argument showing exponential convergence in the first phase, where we assume that the feedforward weights $\mathbf{W}$ remain fixed—that is, $\frac{d\mathbf{W}}{dt} = 0$. Recall the convex Lyapunov function $L_0(\mathbf{W}, \mathbf{M}) = \|\mathbf{W} \mathbf{A} \mathbf{W}^\top - \mathbf{M}^3\|^2$ from section 7. Suppose $\mathbf{W}$ is fixed and $\mathbf{M}(t)$ is a solution of the ODE, equation 2.4. By the product rule and the ODE, equation 2.4, and suppressing the dependence on t , we have

$$\tau \frac{d}{dt} \mathbf{M}^3 = \mathbf{M} \mathbf{W} \mathbf{A} \mathbf{W}^\top \mathbf{M}^{-1} + \mathbf{W} \mathbf{A} \mathbf{W}^\top + \mathbf{M}^{-1} \mathbf{W} \mathbf{A} \mathbf{W}^\top \mathbf{M} - 3\mathbf{M}^3.$$

Next, by the chain rule, the previous display and the cyclic property of the trace operator, we have

$$\begin{aligned} \tau \frac{d}{dt} L_0(\mathbf{W}, \mathbf{M}) &= -2\text{Tr}[(\mathbf{W} \mathbf{A} \mathbf{W}^\top - \mathbf{M}^3)(2\mathbf{M} \mathbf{W} \mathbf{A} \mathbf{W}^\top \mathbf{M}^{-1} + \mathbf{W} \mathbf{A} \mathbf{W}^\top - 3\mathbf{M}^3)] \\ &= -4 \left\| \mathbf{M}^{\frac{1}{2}} (\mathbf{W} \mathbf{A} \mathbf{W}^\top - \mathbf{M}^3) \mathbf{M}^{-\frac{1}{2}} \right\|^2 - 2L_0(\mathbf{W}, \mathbf{M}) \\ &\leq -2L_0(\mathbf{W}, \mathbf{M}). \end{aligned}$$

Therefore, in the first phase, the Lyapunov function $L_0(\mathbf{W}, \mathbf{M})$ converges exponentially to zero.

E.2 Estimation of Invariant Manifolds in the Scalar Setting. Consider the scalar setting $n = k = 1$, in which case the critical points are $\pm(a, a)$. Figure 6 shows the streamlines of the ODE, equations 2.3 and 2.4, for $w > 0$, computed using Mathematica's built-in `Streamline` function, converging for different $\tau \in (0, \frac{1}{2}]$. The invariant manifolds $\mathcal{O}_0$ and $\mathcal{O}_{\frac{1}{2}}$ are shown in orange and green, respectively. For intermediate τ , we numerically estimate $\mathcal{O}_\tau$ using the method described in the next paragraph. Note that $\mathcal{O}_\tau$ appears to interpolate between $\mathcal{O}_0$ and $\mathcal{O}_{\frac{1}{2}}$ as τ varies from 0 to $\frac{1}{2}$, and this is supported by further simulations (not shown).

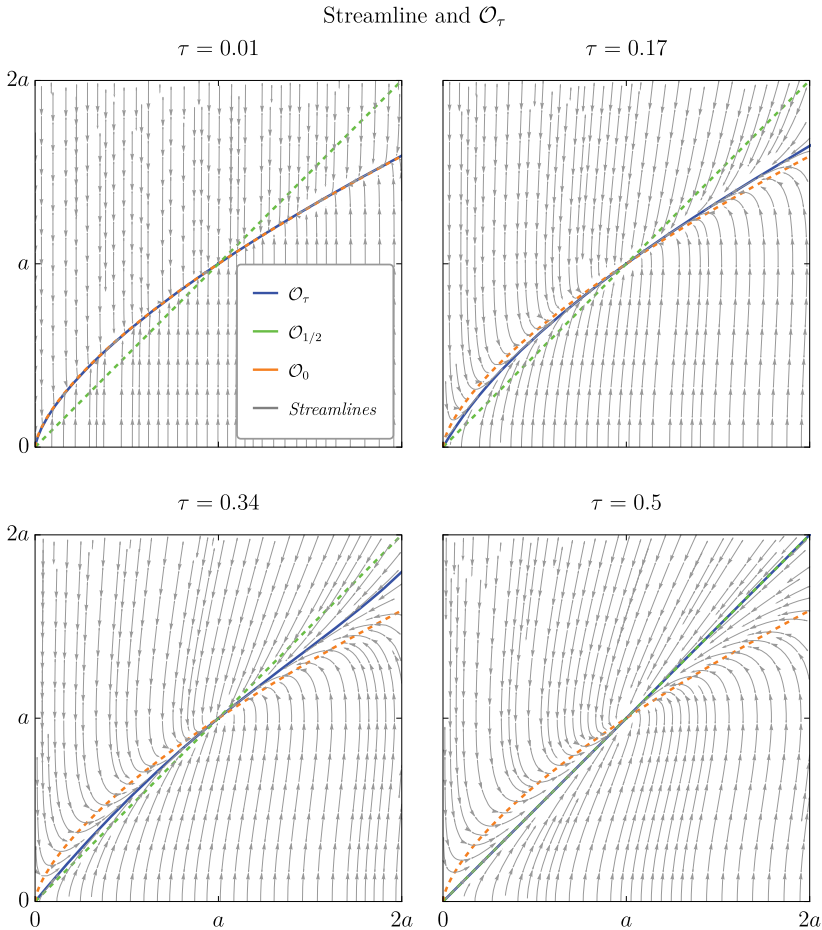


Figure 6: For each τ , the gray arrows show the streamlines for the ODE, equations 2.3 and 2.4 when $n = k = 1$. The solid blue line is a numerical estimate of the invariant set $\mathcal{O}_\tau$. The orange and green dashed lines are the invariant sets $\mathcal{O}_0$ and $\mathcal{O}_{\frac{1}{2}}$, respectively.

E.2.1 Numerical Estimation of $\mathcal{O}_\tau$ for $w > 0$. To estimate $\mathcal{O}_\tau$ when $a < w < 2a$, we define a function $h : (0, \infty)^2 \rightarrow \mathbb{R}^2$ as follows. Given a point (w_0, m_0) , let $(w(t), m(t))$ denote the solution of the ODE with initial condition (w_0, m_0) and set $h(w_0, m_0) = \lim_{t \rightarrow \infty} (\dot{w}(t), \dot{m}(t))$. A point (w_0, m_0) is in $\mathcal{O}_\tau$ if $h(w_0, m_0)$ is parallel to the eigenvector of the Jacobian of the vector field $\mathbf{G}$ with the smallest associated eigenvalue. We then perform a numerical search to find a point $(w_0, m_0) \in \mathcal{O}_\tau$ on the line $w = 2a$ and then compute the solution of the ODE starting at (w_0, m_0) to estimate $\mathcal{O}_\tau$ for $a < w < 2a$. We implemented this procedure with the help of Mathematica's built-in functions

NDSolve and FindRoot. To estimate $\mathcal{O}_\tau$ for $0 < w < a$, we could use a similar argument starting with points (w_0, m_0) near the origin. However, there is an alternative approach that does not require a numerical search. Rather, we assume that the origin is in the closure of the invariant manifold $\mathcal{O}_\tau$ and the invariant manifold can be linearly approximated when m is small; that is, the invariant manifold near the origin is well approximated by $\{(v_0 m, m) : m \approx 0\}$ for some $v_0 > 0$. Under this assumption, when (w_0, m_0) are small, the derivatives of the solutions to the ODE, equations 2.3 and 2.4, at $t = 0$ are approximately $\dot{w}(0) \approx \frac{2aw_0}{m_0}$ and $\dot{m}(0) \approx \frac{aw_0^2}{\tau m_0^2}$. Using the fact that $w_0 \approx v_0 m_0$ and $\dot{w}(0) \approx v_0 \dot{m}(0)$, we see that $2av_0 \approx \frac{a}{\tau} v_0^3$ and so $v_0 = \sqrt{2\tau}$. Therefore, we estimate $\mathcal{O}_\tau$ by computing a solution starting at $(\sqrt{2\tau}\alpha, \alpha)$ for $\alpha > 0$ small. This numerical procedure can also be generalized to estimate the invariant manifolds when $n, k > 1$; however, visualizing the manifolds is more difficult.

References

- Borkar, V. S. (1997). Stochastic approximation with two time scales. *Systems and Control Letters*, 29(5), 291–294. 10.1016/S0167-6911(97)90015-3
- Caron, R., & Traynor, T. (2005). *The zero set of a polynomial*. University of Windsor technical report WMSR 05-03.
- Centorrino, V., Bullo, F., & Russo, G. (2025). *Similarity matching networks: Hebbian learning and convergence over multiple time scales*. arXiv:2506.06134.
- Chou, C.-N., & Wang, M. B. (2020). ODE-inspired analysis for the biological version of Oja's rule in solving streaming PCA. In *Proceedings of the Conference on Learning Theory*, pp. 1339–1343.
- Földiák, P. (1989). Adaptive network for optimal linear feature extraction. In *Proceedings of IEEE/INNS International Joint Conference on Neural Networks*, vol. 1, pp. 401–405.
- Hirsch, M. W., Pugh, C. C., & Shub, M. (1977). *Invariant manifolds*. In *Mathematics*. Springer-Verlag.
- Hornik, K., & Kuan, C.-M. (1992). Convergence analysis of local feature extraction algorithms. *Neural Networks*, 5(2), 229–240. 10.1016/S0893-6080(05)80022-X
- Kushner, H. J., & Clark, D. S. (2012). *Stochastic approximation methods for constrained and unconstrained systems*. Springer Science & Business Media.
- LaSalle, J. P. (1968). Stability theory for ordinary differential equations. *Journal of Differential Equations*, 4(1), 57–65. 10.1016/0022-0396(68)90048-X
- Leen, T. K. (1991). Learning in linear feature-discovery networks. In *Proceedings of Adaptive Signal Processing*, pp. 472–481.
- Leen, T., Rudnick, M., & Hammerstrom, D. (1990). Hebbian feature discovery improves classifier efficiency. In *Proceedings of the 1990 IJCNN International Joint Conference on Neural Networks*, pp. 51–56.
- Lin, T., Jin, C., & Jordan, M. (2020). On gradient descent ascent for nonconvex-concave minimax problems. In *Proceedings of the International Conference on Machine Learning*, pp. 6083–6093.

- Lipshutz, D., Bahroun, Y., Golkar, S., Sengupta, A. M., & Chklovskii, D. B. (2021). A biologically plausible neural network for multichannel canonical correlation analysis. *Neural Computation*, 33(9), 2309–2352. 10.1162/neco_a_01414
- Lipshutz, D., Bahroun, Y., Golkar, S., Sengupta, A. M., & Chklovskii, D. B. (2023). A normative framework for deriving neural networks with multicompartmental neurons and non-Hebbian plasticity. *PRX Life*, 1, 013008. 10.1103/PRXLife.1.013008
- Mirsky, L. (1975). A trace inequality of John von Neumann. *Monatshefte für Mathematik*, 79(4), 303–306.
- Oja, E. (1982). Simplified neuron model as a principal component analyzer. *Journal of Mathematical Biology*, 15(3), 267–273. 10.1007/BF00275687
- Oja, E. (1992). Principal components, minor components, and linear neural networks. *Neural Networks*, 5(6), 927–935. 10.1016/S0893-6080(05)80089-9
- Oja, E., & Karhunen, J. (1983). An analysis of convergence for a learning version of the subspace method. *Journal of Mathematical Analysis and Applications*, 91(1), 102–111. 10.1016/0022-247X(83)90095-1
- Oja, E., & Karhunen, J. (1985). On stochastic approximation of the eigenvectors and eigenvalues of the expectation of a random matrix. *Journal of Mathematical Analysis and Applications*, 106(1), 69–84. 10.1016/0022-247X(85)90131-3
- Pehlevan, C., & Chklovskii, D. B. (2014). A Hebbian/anti-Hebbian network derived from online non-negative matrix factorization can cluster and discover sparse features. In *Proceedings of the 2014 48th Asilomar Conference on Signals, Systems and Computers*, pp. 769–775.
- Pehlevan, C., & Chklovskii, D. B. (2019). Neuroscience-inspired online unsupervised learning algorithms: Artificial neural networks. *IEEE Signal Processing Magazine*, 36(6), 88–96. 10.1109/MSP.2019.2933846
- Pehlevan, C., Hu, T., & Chklovskii, D. B. (2015). A Hebbian/anti-Hebbian neural network for linear subspace learning: A derivation from multidimensional scaling of streaming data. *Neural Computation*, 27(7), 1461–1495. 10.1162/NECO_a_00745
- Pehlevan, C., Sengupta, A. M., & Chklovskii, D. B. (2018). Why do similarity matching objectives lead to Hebbian/anti-Hebbian networks? *Neural Computation*, 30(1), 84–124. 10.1162/neco_a_01018
- Plumbley, M. D. (1995). Lyapunov functions for convergence of principal component algorithms. *Neural Networks*, 8(1), 11–23. 10.1016/0893-6080(95)91644-9
- Potrie, R., & Monzón, P. (2009). Local implications of almost global stability. *Dynamical Systems*, 24(1), 109–115. 10.1080/14689360802474657
- Rubner, J., & Schulten, K. (1990). Development of feature detectors by self-organization. *Biological Cybernetics*, 62(3), 193–199. 10.1007/BF00198094
- Rubner, J., & Tavan, P. (1989). A self-organizing network for principal-component analysis. *EPL (Europhysics Letters)*, 10(7), 693. 10.1209/0295-5075/10/7/015
- Sanger, T. D. (1989). Optimal unsupervised learning in a single-layer linear feed-forward neural network. *Neural Networks*, 2(6), 459–473. 10.1016/0893-6080(89)90044-0